Supply Chain Expert Playbook: Monitoring Every Factor That Signals Drug Price Increases in UK Pharmacies

Executive Summary

As a supply chain expert managing procurement for 30+ UK pharmacies, predicting price increases requires monitoring 57 distinct signals across 4 time horizons — from leading indicators visible 3–12 months before a price rise, to real-time signals that demand immediate purchasing action. This playbook maps every factor to its specific data source, the exact API or feed to capture it, the update frequency required, and how to wire it into an n8n automation workflow. The core insight from 24 years of WaveData market analysis is that price increases are almost always preceded by the Drug Tariff price falling below an identifiable "unprofitable floor" per molecule — and that floor can be calculated from historical data. Once prices drop below that floor, manufacturers stop commissioning stock, supplies dry up within weeks, and price concessions follow inevitably.[1][2]

Part 1: The 4-Tier Early Warning System

The monitoring framework is structured into four tiers, ordered by how far in advance each signal predicts a price increase. For a 30+ pharmacy group, all four tiers should be running simultaneously.

Tier 1: Leading Indicators (3–12 Months Ahead) — 18 Signals

These signals predict price increases long before they materialise. They inform **strategic purchasing decisions**, contract negotiations, and buffer stock planning.

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
1	**Drug Tariff price approaching "unprofitable floor"**	When DT price falls below a molecule's historical shortage-trigger price, a shortage and price spike is 3–6 months away	WaveData historical market prices (24 years of data)[2]	WaveData subscription API or manual extract	HTTP Request (scheduled monthly)
2	**Patent expiry calendar**	Upcoming patent cliffs = price drops; but also = originator "product hopping" and supply disruption	DrugPatentWatch.com database[3]	DrugPatentWatch API	HTTP Request (monthly)
3	**NICE appraisal pipeline**	New NICE approvals = new demand; rejections = demand stays with alternatives	NICE website / API feed[4]	RSS Feed / HTTP Request	HTTP Request (weekly)
4	**DaSH portal notifications (6-month advance warning)**	MAHs legally required to report supply issues 6 months in advance to DHSC[5][6]	DHSC DaSH portal	Freedom of Information / MHRA alerts	Webhook (as available)
5	**Manufacturer**	Company pulling a	DaSH portal +	SPS websit	HTTP Requ

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
	discontinuation notices	product from UK market = supply gap in 6–12 months	SPS Medicines Supply Tool[5][7]	e scraping / alerts	est (daily)
6	**Generic manufacturer count per molecule**	Declining number of suppliers = rising shortage risk; ≤2 suppliers = high risk	Drug Tariff Part VIIIA supplier data[8]	NHSBSA Drug Tariff PDF parsing	Code Node (monthly)
7	**API source country risk assessment**	Geopolitical tension in India/China = API supply risk 6–12 months later	Geopolitical risk indices + news sentiment[9][10]	News API + LLM sentiment analysis	AI Agent Node (daily)
8	**GBP exchange rate trends**	Sustained GBP depreciation vs INR/CNY = import cost increases flowing through in 3–6 months	Bank of England, ECB, forex APIs[11]	Exchange Rate API (e.g., exchangerate-api.com)	HTTP Request (daily)
9	**Energy price futures**	Rising European energy futures = manufacturing cost increases in 3–6 months	Ofgem, ICE energy futures[12]	Commodities API[13]	HTTP Request (daily)

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
10	**Raw material commodity prices**	HDPE, glass, petrochemical input cost trends	ChemAnalyst, Tradefeeds API[14][15]	Commodity price API	HTTP Request (weekly)
11	**VPAG rebate rate changes**	Reduction from 22.9% → 14.5% means branded drugs get more expensive for NHS	DHSC VPAG publications[4]	Gov.uk RSS / manual monitoring	HTTP Request (quarterly)
12	**US tariff announcements on pharma**	US tariffs disrupt global API supply chains, redirect trade flows	White House / USTR announcements[16]	News API + keyword alerts	AI Agent Node (real-time)
13	**International shipping disruption indices**	Red Sea, Suez Canal, port congestion = 12+ day delays on API shipments	Freightos Baltic Index, Drewry Container Index	Shipping rate API	HTTP Request (weekly)
14	**NHS budget allocation decisions**	Funding cuts to medicines budgets = procurement pressure = shortages	King's Fund, NHS England publications[17]	RSS / web scraping	HTTP Request (quarterly)

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
15	**Demographic projections**	Ageing population data projecting chronic disease burden increases	ONS population projections[18]	ONS API (beta)	HTTP Request (annually)
16	**Clinical trial results for new indications**	Successful Phase III = upcoming demand surge for existing drug or therapeutic area	ClinicalTrials.gov, EudraCT	ClinicalTrials.gov API	HTTP Request (monthly)
17	**Biosimilar pipeline for reference biologics**	Upcoming biosimilar entry = price down; delays = price stays high	EMA biosimilar tracking[19]	EMA website monitoring	HTTP Request (quarterly)
18	**UK-EU regulatory divergence tracker**	Further divergence = more supply barriers = higher costs	MHRA vs EMA comparison[20]	Manual tracking / industry publications	HTTP Request (quarterly)

Tier 2: Confirming Indicators (1–3 Months Ahead) — 15 Signals

These signals confirm that a price movement is building. They inform **tactical purchasing adjustments** — increasing order quantities, switching suppliers, or accelerating purchases.

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
19	**Drug Tariff monthly price changes**	DT price cut this month may trigger manufacturer withdrawal next month	NHSBSA Drug Tariff PDF/API[21][22]	NHSBSA Open Data Portal API[23]	HTTP Request (1st of month)
20	**Wholesaler "out of stock" frequency**	Rising OOS reports from AAH/Alliance Healthcare = shortage building[24]	Wholesaler portals + pharmacist reports	Internal pharmacy data / CPE reports	Webhook (daily)
21	**Wholesaler delivery lead time changes**	Same-day becoming next-day or 2-day = supply tightening	Wholesaler order systems[25]	Internal ERP / wholesaler API	Code Node (daily)
22	**CPE price concession request volume**	Spike in requests = many pharmacies cannot buy at DT price	Community Pharmacy England website[26][27]	CPE website scraping	HTTP Request (weekly)
23	**Prescription volume spike detection**	Sudden increase in Rx volume for a drug = demand surge building	OpenPrescribing API[28][29]	`/api/1.0/spending/?code={BNF_code}`	HTTP Request (monthly)

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
24	**Weather forecast (14-day)**	Cold snap → respiratory drug demand; heatwave → cardiovascular	OpenWeather One Call API 3.0[30]	OpenWeather API (temp, AQ, pollen, humidity)	HTTP Request (hourly)
25	**Flu/RSV surveillance data**	Rising infection rates = demand surge for antivirals, antibiotics in 2–4 weeks	PHE/UKHSA weekly flu reports[31]	UKHSA surveillance API / reports	HTTP Request (weekly)
26	**Pollen count forecast**	High pollen = antihistamine demand surge in 1–2 weeks	Met Office / Ambee pollen API[32][33]	Ambee Pollen API / OpenWeather	HTTP Request (daily)
27	**Social media demand signals**	TikTok/Instagram influencer promoting weight-loss drugs = demand spike	Twitter/X API, TikTok analytics[34][35]	Social media API + LLM sentiment	AI Agent Node (daily)
28	**Google Trends health search volume**	Rising searches for conditions = prescriptions	Google Trends API	Unofficial Google Trends API	Code Node (weekly)

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
		following in 2–4 weeks		(pytrends)	
29	**NHS 111 call volume by condition**	Surge in calls about specific symptoms = GP prescriptions following	NHS 111 data[31]	NHS Digital API	HTTP Request (weekly)
30	**Competitor pharmacy pricing intelligence**	Other pharmacies raising prices = supply tightening market-wide	WaveData market price tracker, Dispex	WaveData API[2]	HTTP Request (weekly)
31	**Wholesaler price list changes**	Price increases from wholesalers = confirmed cost pressure	AAH, Alliance, Phoenix price lists	Wholesaler EDI feed / API	Webhook (as published)
32	**Seasonal calendar triggers**	December = annual prescription spike; September = school return infections	Calendar logic[36]	Built-in date logic	Schedule Node
33	**Financial year-end procurement**	March procurement rush = temporary price inflation	Calendar + historical patterns[36]	Built-in date logic	Schedule Node

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
	patterns				

Tier 3: Real-Time Signals (Days to Weeks) — 14 Signals

These signals require **immediate action** — buy now, switch supplier, or alert pharmacy managers.

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
34	**Price concession published**	Official confirmation that Drug Tariff price is unviable; market price already higher	CPE website[26], BSONI[37]	CPE concession page scraping	HTTP Request (daily, 3x/month)
35	**Serious Shortage Protocol (SSP) activated**	DHSC has formally declared a shortage; pharmacists can substitute[38]	NHSBSA SSP notifications	NHSBSA alerts	Webhook
36	**MHRA supply disruption alert**	MHRA-issued alert = confirmed national supply problem	MHRA website / Central Alerting System[39]	CAS email alerts / RSS	Webhook
37	**Your pharmacies' own purchase price vs DT**	When actual purchase price > Drug Tariff price = losing money on every dispense	Internal pharmacy ERP system	Database query	Code Node (per order)
38	**Wholesaler allocation/rationi**	Wholesaler limiting quantities per pharmacy =	Wholesaler communication / order	Wholesaler EDI / internal	Webhook

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
	ng initiated	severe shortage	rejection	reporting	
39	**Multiple wholesalers reporting OOS simultaneously**	Cross-wholesaler OOS = genuine market shortage, not single-supplier issue	Multi-wholesaler order system check[5]	Parallel queries to wholesaler APIs	HTTP Request (daily)
40	**Rapid GBP/INR or GBP/CNY movement**	>2% daily move = import cost shock flowing through immediately	Forex API real-time	Exchange Rate API with alerting	HTTP Request + IF Node (hourly)
41	**Major manufacturer quality recall**	Recall = immediate supply gap for that manufacturer's volume	MHRA recall alerts	MHRA alerts API / RSS	Webhook
42	**Geopolitical event alert**	Export ban, factory closure, shipping lane disruption	Datamin r Pulse / news sentiment[10]	News API + LLM classification	AI Agent Node (real-time)
43	**Extreme weather event**	Storm/flood disrupting distribution; heatwave creating demand surge	Met Office severe weather alerts[30]	Met Office API	HTTP Request + IF Node

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
44	**COVID/flu/RSV outbreak escalation**	Pandemic-level demand surge (e.g., March 2020: 93M prescriptions)	UKHSA alert levels[40]	UKHSA API / RSS	Webhook
45	**Port/logistics disruption**	Strike, congestion, or closure at key ports (Felixstowe, Dover, Rotterdam)	Shipping industry alerts[16]	Port authority feeds / news	AI Agent Node
46	**Cyber attack on pharma/logistics**	Ransomware or attack disrupting manufacturer or wholesaler systems	Cyber threat intelligence[10]	News alerts + NCSC feeds	Webhook
47	**Buffer stock depletion at your pharmacies**	Internal stock for critical medicines falling below safety threshold	Internal pharmacy stock management	Database monitoring query	Code Node (daily)

Tier 4: Lagging Confirmation (Already Happening) — 10 Signals

These signals confirm a price increase has occurred. They inform **retrospective analysis**, model retraining, and future prediction improvement.

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
48	**Drug Tariff price increased this month**	Official confirmation of price rise in reimbursement	NHSBSA Drug Tariff[21]	NHSBSA Open Data Portal[23]	HTTP Request (monthly)
49	**Price concession ended (price stays elevated)**	Post-concession DT price higher than pre-shortage = permanent shift	Drug Tariff historical data[1]	OpenPrescribing Drug Tariff data[29]	HTTP Request (monthly)
50	**Actual cost per item in NHSBSA data**	Real-world cost data 2 months lagged	NHSBSA Prescription Cost Analysis[41][23]	OpenPrescribing /api/1.0/spending/	HTTP Request (monthly)
51	**OpenPrescribing cost trend analysis**	Spending per item trending upward over 3+ months	OpenPrescribing trend tools[42][29]	OpenPrescribing API	HTTP Request (monthly)
52	**Manufacturer list price increase filed**	Originator/branded medicine list price increase	DHSC pricing notifications	Gov.uk publications	HTTP Request (monthly)

#	Signal	What It Tells You	Data Source	How to Get It	n8n Node
53	**Competitor pharmacy margin analysis**	If competitors are losing money on a drug, whole market is under pressure	WaveData market intelligence[2]	WaveData reports	HTTP Request (monthly)
54	**NHS hospital dispensing cost reports**	Hospital procurement cost trends (3-6 month lag)	NHS England / GIRFT data	NHS Digital datasets	HTTP Request (quarterly)
55	**Parallel import volume decline**	Fewer parallel imports = less competition = higher prices (post-Brexit effect)[20]	MHRA parallel import data	MHRA database	HTTP Request (quarterly)
56	**Annual PCA statistics**	Year-on-year cost and volume trends confirmed	NHSBSA Prescription Cost Analysis [23]	NHSBSA Open Data Portal	HTTP Request (annually)
57	**Model prediction vs actual price (feedback loop)**	Prediction error analysis improves future accuracy	Internal AI model performance	Model monitoring dashboard	Code Node (monthly)

Part 2: The Critical Data Sources — Complete API & Feed Directory

Primary Data Sources (Must-Have for Day 1)

Data Source	What You Get	API/Access Method	Cost	Update Frequency	Priority
NHSBSA Open Data Portal	Drug Tariff prices, prescription volumes, costs, pharmacy data	REST API + CSV downloads[23]	Free	Monthly	Critical
OpenPrescribing	5 years of prescribing data, Drug Tariff prices, price concessions, spending by practice/ICB	REST API (/api/1.0/spending/, /api/1.0/spending_by_org/)[28][29]	Free	Monthly (2-month lag)	Critical
Drug Tariff PDF	Official monthly reimbursement prices for all Part VIII drugs	NHSBSA website PDF[43]	Free	Monthly (1st of month)	Critical
CPE Price Concessions	Active price concessions with drug name, pack size, concessionary price	CPE website scraping[26]	Free	3x/month (1st, 8th, 15th)	Critical
BSO NI Concessiona	Northern Ireland-specific concession prices	BSO website[37]	Free	Monthly	Critical (for NI)

Data Source	What You Get	API/Access Method	Cost	Update Frequency	Priority
ry Prices					
dm+d (Dictionary of Medicines)	Product codes, formulations, strengths, pack sizes, manufacturers	NHS Digital TRUD download	Free	Weekly	Critical
Wave Data	24 years of generic market prices, shortage trigger analysis, competitor pricing	Subscription service[2]	Paid (£2K-5K/yr est.)	Weekly	High
Exchange Rate API	GBP/EUR, GBP/USD, GBP/INR, GBP/CNY real-time rates	exchangerate-api.com or Open Exchange Rates	Free tier available	Real-time	High

Secondary Data Sources (Add in Phase 2)

Data Source	What You Get	API/Access Method	Cost	Update Frequency
OpenWeather One Call 3.0	Temperature, humidity, air quality (PM2.5, ozone, NO2), pollen by type, UV index	REST API with location parameter[30]	Free tier (1000 calls/day)	Hourly
Meteostat API	Historical weather data (used in NHS XGBoost study, R^2=0.80)	REST API[44]	Free	Historical + daily
Ambee Pollen API	Detailed pollen data by species, correlated with antihistamine demand	REST API[32][33]	Free tier available	Daily
UKHSA Flu Surveillance	Weekly flu/RSV incidence rates, GP consultation rates	PDF reports / syndromic surveillance API[31]	Free	Weekly
DrugPatentWatch	Patent expiry dates, generic entry opportunity dates for UK	Website + data downloads [3]	Paid subscription	Monthly
ChemAnalyst	Pharma API prices, excipient prices, raw material cost trends	Platform + API[14]	Paid	Daily
Commodities API	Oil, natural gas, HDPE,	REST API (JSON)[13]	Free tier (100	Real-time

Data Source	What You Get	API/Access Method	Cost	Update Frequency
	polyethylene, copper, steel prices		calls/month)	
Google Trends (pytrends)	Health condition search volume trends as demand proxy	Unofficial Python library	Free	Weekly
NHS 111 Pathways	Call volumes by symptom group, regional breakdown	NHS Digital	Free (public data)	Weekly

Tertiary Data Sources (Phase 3 — Advanced Intelligence)

Data Source	What You Get	API/Access Method	Cost
MHRA / DaSH Portal	Manufacturer shortage/discontinuation reports (legally required 6 months ahead)[6]	FOI requests / MHRA alerts	Free
SPS Medicines Supply Tool	NHS England medicine shortage alerts and management guidance[45][5]	SPS website monitoring	Free
EU FMD/EMVS Repository	Pack-level serialisation data showing supply vs dispense rates[46]	NCA access (complex)	Institutional
Freightos Baltic Index	Container shipping rates (proxy for logistics costs)	Freightos API	Paid
News API + LLM Sentiment	Real-time geopolitical, trade, and pharma industry news analysis[10]	NewsAPI.org + OpenAI	~$50/month
Social Media Monitoring	TikTok/Instagram/Reddit demand signals for weight-loss, ADHD, etc.[34]	Platform APIs + LLM	Variable

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Part 3: The Price Increase Signal Scoring Model

How to Score and Prioritise Alerts

Each signal from the 57 monitoring points should be scored on three dimensions to determine action urgency:

Dimension	Weight	Scale	Description
Probability	40%	1–5	How likely is this signal to result in a price increase?
Impact	35%	1–5	How large would the price increase be? (£ and % terms)
Velocity	25%	1–5	How quickly will the price increase materialise?

Composite Risk Score = (Probability × 0.40) + (Impact × 0.35) + (Velocity × 0.25)

Action thresholds:

Score	Alert Level	Action
4.0–5.0	**RED — Buy Now**	Immediate bulk purchase across all 30+ pharmacies; maximum buffer stock
3.0–3.9	**AMBER — Accelerate**	Bring forward next order cycle; negotiate volume discount; diversify supplier
2.0–2.9	**YELLOW — Watch**	Monitor weekly; no immediate action; prepare contingency
1.0–1.9	**GREEN — Normal**	Continue routine purchasing; log for trend analysis

Example: Atorvastatin Shortage Signal Scoring (February 2026)

Signal	Probability	Impact	Velocity	Score
Drug Tariff price near floor (£1.27)	5	4	4	4.40
Manufacturer exit from UK market[1]	5	5	3	4.50
Wholesaler OOS reports increasing[47]	5	4	5	4.75
Price concession active	5	3	5	4.30
Composite for Atorvastatin	**5.0**	**4.0**	**4.25**	**4.49**

This would trigger an immediate "BUY NOW" alert across all 30 pharmacies.[2][47]

Part 4: The WaveData "Unprofitable Floor" Method

WaveData's 24-year analysis reveals the single most powerful predictor of generic medicine price increases: **when the market price falls below the molecule-specific "unprofitable floor," a shortage and price spike follows within 3–6 months**.[2]

How the Method Works

For each generic molecule, historical data reveals a price point below which manufacturers consistently withdraw from the market:

1. **Collect historical price data** for each molecule (minimum 5 years, ideally 10+)
2. **Identify all historical shortage events** (price concessions as proxy)

3. **Find the lowest market price that preceded each shortage** (the "trigger price")
4. **Calculate the "unprofitable floor"** as the lowest trigger price that preceded shortages on ALL occasions

Example: Losartan 12.5mg 28 tablets[2]

Shortage Event	Preceding Trigger Price	Shortage Triggered?
Event 1	£1.64	✅ Yes
Event 2	£2.86	✅ Yes (but £2.86 didn't always cause shortages)
Event 3	£2.11	✅ Yes
Event 4	£1.65	✅ Yes
Unprofitable floor	**£1.64**	**100% predictive**

For atorvastatin: The CCA analysis shows the Drug Tariff price was cut to £0.89 in 2022, which triggered manufacturer exit, shortages, and eventual price concessions costing the NHS millions. The price was subsequently reset to £1.64 in January 2024. The "unprofitable floor" for atorvastatin is approximately £0.89–£1.00.[1]

Scaling This Across 30+ Pharmacies

For your stock module, build a database of unprofitable floor prices for the **top 200 generic molecules by dispensing volume** across your 30+ pharmacies. The n8n workflow would:

1. **Ingest monthly Drug Tariff prices** via NHSBSA API
2. **Compare each molecule's DT price against its unprofitable floor**
3. **Flag molecules where DT price is within 15% of the floor** (early warning)
4. **Trigger buying alerts when DT price crosses the floor** (buy immediately before shortage hits)

Part 5: n8n Workflow Architecture

Workflow 1: Daily Monitoring Pipeline

```
Schedule Trigger (6:00 AM daily)
  → HTTP Request: Check CPE price concessions page
  → HTTP Request: Check BSO NI concessionary prices
  → HTTP Request: Fetch GBP exchange rates (INR, CNY, EUR, USD)
  → HTTP Request: Fetch weather/pollen data (OpenWeather)
  → HTTP Request: Check MHRA shortage alerts
  → Code Node: Score all signals against threshold matrix
  → IF Node: Score ≥ 3.0?
    → Yes: Send Slack/Teams alert to procurement team
    → Yes: Update Airtable/database with alert details
    → No: Log to daily monitoring spreadsheet
```

Workflow 2: Monthly Price Analysis Pipeline

```
Schedule Trigger (2nd of each month)
  → HTTP Request: Download new Drug Tariff PDF from NHSBSA
  → Code Node: Parse PDF, extract Part VIIIA prices
  → HTTP Request: Fetch OpenPrescribing spending data for top 200 molecul
  → Code Node: Compare DT prices vs unprofitable floor database
  → Code Node: Calculate month-on-month price changes
  → Code Node: Identify molecules with ≥3 consecutive DT price cuts
  → HTTP Request: Fetch patent expiry data for flagged molecules
  → AI Agent: Summarise risk assessment in natural language
  → Send report to procurement team
  → Update prediction model with new data
```

Workflow 3: Real-Time Shortage Response

```
Webhook Trigger (incoming wholesaler OOS notification)
  → Code Node: Check if molecule is already flagged
  → HTTP Request: Query all wholesaler APIs for cross-market availability
  → IF Node: OOS across ≥2 wholesalers?
```

```
  → Yes: 🔴 RED alert — immediate buying action
  → Yes: HTTP Request: Check alternative suppliers/buying groups
  → Yes: Send urgent alert to all 30+ pharmacy managers
  → No: 🟢 Log and monitor
```

Workflow 4: Demand Forecasting Pipeline

```
Schedule Trigger (Weekly, Monday 7:00 AM)
  → HTTP Request: Fetch weather forecast (14-day)
  → HTTP Request: Fetch UKHSA flu surveillance data
  → HTTP Request: Fetch pollen forecast (Ambee API)
  → HTTP Request: Fetch Google Trends health searches
  → Code Node: Run XGBoost demand prediction model
   (trained on NHS prescription data + weather, R²=0.80)
  → Code Node: Flag molecules where predicted demand > current stock
  → Send demand forecast report to procurement team
```

Part 6: The 850 Million Pack Problem

Understanding the systemic risk is critical. Pharmacy industry leaders have warned that **over 850 million generic medicine packs are reimbursed at 99p or less for a month's supply** — a level described as "simply not sustainable". This means the vast majority of high-volume generics in your portfolio are at permanent risk of the unprofitable-floor dynamic.[48][49]

The retained margin for community pharmacy stands at £900 million, but price erosion continues. For a 30+ pharmacy group, the strategic implication is clear: **monitoring is not optional — it's existential**. The pharmacies that survive will be those with superior intelligence on when prices are about to move, enabling them to:[48]

- Buy ahead of shortages (saving 50–800% vs concession prices)[37]
- Negotiate better terms with multiple wholesalers using volume as leverage[36]
- Diversify supplier base to avoid single-wholesaler dependency[5]

- Time purchases to avoid December and March price spikes[36]
- Build buffer stock for high-risk molecules identified by the unprofitable floor analysis[2]

Quick-Start Implementation Plan

Week	Action	n8n Workflow
1	Set up NHSBSA Drug Tariff PDF auto-download + parsing	Workflow 2 (partial)
2	Connect OpenPrescribing API for top 200 molecules by volume	Workflow 2
3	Set up CPE price concession scraping (3x/month)	Workflow 1
4	Connect exchange rate + commodity price APIs	Workflow 1
5	Build unprofitable floor database for top 200 generics	Workflow 2 (Code Node)
6	Connect OpenWeather + UKHSA flu data	Workflow 4
7	Build signal scoring engine + alert thresholds	Workflow 1 (Code Node)
8	Deploy real-time wholesaler OOS monitoring	Workflow 3
9	Train initial XGBoost demand prediction model	Workflow 4
10	Full system integration testing across all 30+ pharmacies	All workflows
11	Add social media + news sentiment monitoring	Workflow 1 (AI Agent)
12	Go live with complete monitoring dashboard	All workflows

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